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MACHINE LEARNING



Now that you understand the ideas behind many fundamental algorithms, we can turn to more advanced ideas. In this chapter, we explore machine learning. *Machine learning* refers to a broad range of methods, but they all share the same goal: finding patterns in data and using them to make predictions. We'll discuss a method called *decision trees* and then build one that can predict a person's level of happiness based on some of their personal characteristics.

Decision Trees

Decision trees are diagrams that have a branching structure resembling a tree. We can use decision trees in the same way we use flowcharts—by answering yes/no questions, we are guided along a path that leads to a final decision, prediction, or recommendation. The process of creating a decision